

closure_evidence_20260926

BOB-249 closure evidence

Revision: 1 **Last modified:** 2026-09-26T15:08:22Z

Command: for t in

```
test_generate_markdown_exports_content_staleness
test_export_staleness_oracle test_export_staleness_parser_edges
test_compute_badges_export_scope
test_workable_items_export_deterministic
test_generate_markdown_exports_engine_parity; do nice -n 19 bash
tests/unit/$t.sh 2>&1 | grep -oE '[0-9]+ passed, [0-9]+ failed' |
tail -1; done | awk '{p+=$1; f+=$3} END{print p" passed, "f"
failed"}'
```

Result Summary: 77 passed, 0 failed **Evidence Layer:** runtime **Test Type:** unit, integration

What was reported

The pre-build invariant 30 (CM-BASH-UNIT-TESTS-EXECUTED) failed with “N tracked file(s) mtime-moved while the bash suite ran”: tracked export twins (.docx/.html/.pdf beside .md) were rewritten during the sweep. Reproduced at the merge-base 120fd78 (15 files) and at main 88767fc (18 files), so it predates the zero-shortcomings audit feature.

Root cause (measured, not inferred)

1. git checkout writes x.docx and x.html before x.md, so after a checkout hundreds of .md files are strictly newer (ns) than their own twins: 226 of 455 md newer than their .docx in a fresh clone.
2. scripts/generate_markdown_exports.sh decided staleness with \$md -nt \$twin (mtime), while the pre-build gate CM-MARKDOWN-EXPORT-SYNC used a content-history oracle.

3. `scripts/compute-badges.sh` ended every run by calling that generator with `NO` path argument, sweeping the whole real tree even when invoked on fixtures; `tests/unit/test_compute_badges_carrier_match.sh` did exactly that inside invariant 30's window (bisect with the trigger present: 42 twins rewritten, 91 files past the marker).
4. `pandoc` stamps the wall-clock time in `docx docProps/core.xml`, so every regeneration was byte-different. The Go export in `scripts/workable-items-export.sh` re-renders the four tracker documents on every run.

Fix (commits afe5d1f a630935 74e5b6f e47346b 624a8d3 5f1c4bc 09aed1f 96767c7)

Staleness now uses the gate's oracle (extended to `.docx`, NUL-safe, full-history); `SOURCE_DATE_EPOCH` pins make output byte-stable; `compute-badges` exports only the in-repo files it modified; the generator installs atomically in the twin's directory, heals structurally corrupt twins, fails loudly on an HTML-leg failure, and renders `docs_chain-owned` files exactly as the engine does. Every change was written test-first (RED observed), and two independent reviews (plus scoped re-reviews) approved the final code.

Evidence

Re-runnable command above (sum of the regression tests guarding the root causes; identical output on two consecutive runs, 77 passed, 0 failed, tree clean afterwards).

Fresh-checkout measurement (scratch clone, generator over all 456 in-scope `.md` files): before the fix 310 twins rewritten (42 html, 42 pdf, 226 docx); after 0, with a control needle (a real content edit still regenerates its 3 twins).

Full pre-build sweep, captured from the real run logs, on a quiet tree:

Baseline at merge-base 120fd78 (old constitution pin 25980c1):

```
FAIL [1]: CM-BASH-UNIT-TESTS-EXECUTED: 15 tracked file(s) mtime-  
moved while the bash suite ran: README.docx README.html README.pdf  
docs/Fixed.docx docs/Fixed.html docs/Fixed.pdf docs/  
Fixed_Summary.d
```

```
FAIL [2]: CM-GATE-LEDGER-RATCHET: exit 1 — CM-GATE-LEDGER-  
RATCHET: FAIL — see LEDGER-FAIL lines above; land the missing
```

gate(s), register their deferral(s) against a tracked item, or
(for va

FAIL [3]: CM-GITIGNORE-SWALLOW-GUARD: exit 1 – a first-party
source file is silently swallowed by .gitignore

=== Result: 52 passed, 3 failed ===

main at 88767fc before the fix:

CM-BASH-UNIT-TESTS-EXECUTED: 18 tracked file(s) mtime-moved while
the bash suite ran

=== Result: 53 passed, 2 failed ===

main after the fix, first sweep (tracker re-exported once onto the
stable-date scheme):

PASS [31]: CM-BASH-UNIT-TESTS-EXECUTED: 84 bash unit/pre_build
test(s) green (5 quarantined, 0 expected-red), no-trace verified
across 5903 tracked paths (needle proven)

FAIL [1]: CM-GATE-LEDGER-RATCHET: exit 1 – CM-GATE-LEDGER-
RATCHET: FAIL – see LEDGER-FAIL lines above; land the missing
gate(s), register their deferral(s) against a tracked item, or
(for va

=== Result: 54 passed, 1 failed ===

main after the final code (commit 96767c7), second sweep:

PASS [17]: CM-MARKDOWN-EXPORT-SYNC: all in-scope docs have
fresh .html/.pdf siblings

PASS [26]: docs_chain engine: verify --all in-sync (2 contexts
checked)

PASS [31]: CM-BASH-UNIT-TESTS-EXECUTED: 88 bash unit/pre_build
test(s) green (5 quarantined, 0 expected-red), no-trace verified
across 5908 tracked paths (needle proven)

FAIL [1]: CM-GATE-LEDGER-RATCHET: exit 1 – CM-GATE-LEDGER-
RATCHET: FAIL – see LEDGER-FAIL lines above; land the missing
gate(s), register their deferral(s) against a tracked item, or
(for va

=== Result: 54 passed, 1 failed ===

The tree was clean (0 modified files) after both post-fix sweeps; before
the fix each sweep left 14 to 18 rewritten twins.

Honest boundary

- The remaining sweep failure is CM-GATE-LEDGER-RATCHET
(BOB-237, unimplemented 414 vs baseline 403), unrelated and pre-
existing.

- The sweep lines above are pasted from run logs; only the regression-test command is cheap enough to re-run as the machine-checkable part of this closure.
- Known limits tracked in BOB-250: an owned twin cannot converge after a whitespace-only commit; a quoted docs_chain node path containing a space is mis-parsed; the sha-tag oracle proposal for that residual case is not implemented.